FINITENESS OF THE CLOSED PART Z OVER T WHEN F IS SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Finiteness implies universal closedness. Morphisms of Spaces, lemma finite-proper (cross-chapter, named). Z -> T universally closed.
- Closedness permanence. Morphisms of Spaces lemmas base-change-separated and universally-closed-permanence (cross-chapter, named). Z closed in T x_Y X once f separated.

Source reference. Original equation 04PC: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1519–1519. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KL: focused-seeds/geometry/S9-smg-finite-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NZ: focused-seeds/geometry/S9-smg-finite-separated/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NX: focused-seeds/geometry/S9-smg-finite-separated/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let T be an algebraic space over S, and let (a, Z) be a pair as in 04PC. If f is separated, then Z is closed in $T \times_Y X$.*

Proof. A finite morphism of algebraic spaces is universally closed by Morphisms of Spaces, Lemma 04NZ. Since f is separated so is the morphism $T \times_Y X \to T$, see Morphisms of Spaces, Lemma 03KL. Thus the closedness of Z follows from Morphisms of Spaces, Lemma 04NX. □

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.